SECTION 8.E

Limits and Discontinuities

Limits describe the behavior of a function as the input approaches a particular value. Instead of focusing on the function's exact value at a point, limits examine what the function approaches near that point. In this section, you will evaluate limits graphically and algebraically and classify different types of discontinuities.

Learning Objectives

- I can evaluate one-sided and two-sided limits from graphs.
- I can classify discontinuities as removable, infinite, or jump discontinuities.
- I can evaluate limits algebraically by factoring and simplifying rational expressions.
- I can write limit notation correctly and interpret function behavior near undefined points.

Introduction: Behavior Near a Point

A limit answers the question: what value does the function approach as the input approaches a specific number? The function does not need to be defined at that number for a limit to exist.

Three main behaviors may occur:

- The function approaches a single value (limit exists).
- The left and right sides approach different values (limit does not exist).
- The function increases or decreases without bound (infinite discontinuity).

Real-World Connection

Limits are used in physics to describe instantaneous velocity and acceleration. Engineers analyze system stability by examining behavior near critical thresholds. Economists evaluate how markets behave as prices approach equilibrium.

Historical Note

The formal definition of limits was developed in the 19th century to make calculus rigorous. Mathematicians such as Cauchy and Weierstrass established precise foundations that support modern analysis.

Key Vocabulary and Concepts

Limit: The value a function approaches as the input approaches a given number.

One-Sided Limit: A limit taken from the left (

$$x \to a^-$$

) or right (

$$x \to a^+$$

).

Two-Sided Limit: A limit that exists only if both one-sided limits are equal.

Removable Discontinuity: A hole caused by a canceled factor.

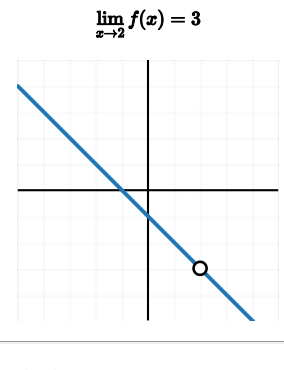
Infinite Discontinuity: A vertical asymptote where the function increases or decreases without bound.

Jump Discontinuity: A break where one-sided limits are not equal.

Core Strategy 1: Evaluate Limits Graphically

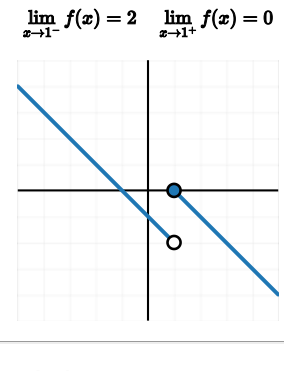
Example 1: Removable Discontinuity

$$\lim_{x\to 2} f(x) = 3$$



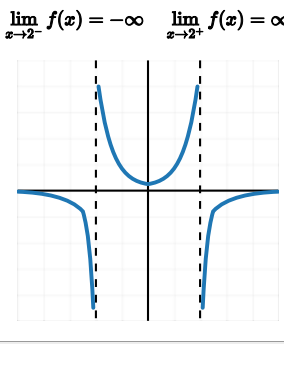
Example 2: Jump Discontinuity

$$\lim_{x\to 1^-} f(x) = 2 \quad \lim_{x\to 1^+} f(x) = 0$$



Example 3: Infinite Discontinuity

$$\lim_{x\to 2^-} f(x) = -\infty \quad \lim_{x\to 2^+} f(x) = \infty$$



Core Strategy 2: Evaluate Limits Algebraically

If substitution produces

$$\frac{0}{0}$$

, factor and simplify before substituting again.

Example 4: Direct Substitution

$$\lim_{x\to 3}(2x^2 - x + 1)$$

Example 5: Indeterminate Form (Factoring Required)

$$\lim_{x\to 2} \frac{x^2-4}{x-2}$$

Example 6: Infinite Limit (Vertical Asymptote)

$$\lim_{x\to 2} \frac{1}{x-2}$$

Classifying Discontinuities

Discontinuities can be identified directly from algebraic structure.

- If a factor cancels → **Removable discontinuity (hole)**.
- If a factor does not cancel and remains in the denominator → **Infinite discontinuity (vertical asymptote)**.
- If a piecewise function changes values abruptly → **Jump discontinuity**.

Example 7:

$$f(x) = \frac{(x-3)(x+2)}{x-3}$$

Summary of Key Concepts

- A limit describes behavior near a point, not necessarily the value at the point.
- If left and right limits are equal, the two-sided limit exists.
- If substitution produces

$$\frac{0}{0}$$

  , factor and simplify.
- If the denominator remains zero and does not cancel, analyze one-sided behavior.
- Discontinuities can be removable, infinite, or jump.

Strategy Checklist

1. Try direct substitution first.
2. If you get

$$\frac{0}{0}$$

   , factor and simplify.
3. If denominator remains zero, analyze left and right behavior.
4. Compare one-sided limits.
5. Classify the discontinuity.

Common Mistakes and How to Avoid Them

1️⃣ Confusing Function Value with Limit

The limit depends on approaching behavior, not the actual function value at the point.

2️⃣ Stopping After Getting 0/0

An indeterminate form means further simplification is required.

3️⃣ Ignoring One-Sided Limits

If left and right limits differ, the two-sided limit does not exist.

4️⃣ Forgetting to Classify the Discontinuity

Every non-continuous point should be identified as removable, infinite, or jump.

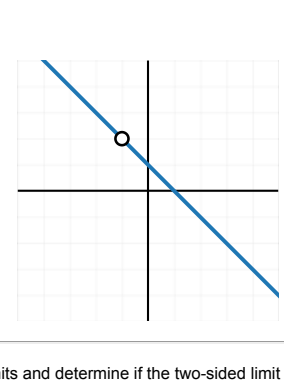
Practice Problems

Graphical Limits

1. Evaluate:

$$\lim_{x\to -1} f(x)$$

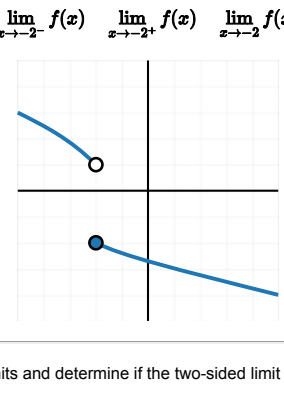
Does the limit exist?



---

2. Evaluate the one-sided limits and determine if the two-sided limit exists:

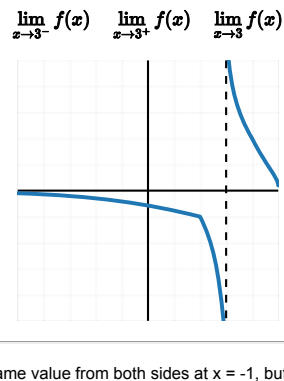
$$\lim_{x\to -2^-} f(x) \quad \lim_{x\to -2^+} f(x) \quad \lim_{x\to -2} f(x)$$



---

3. Evaluate the one-sided limits and determine if the two-sided limit exists:

$$\lim_{x\to 3^-} f(x) \quad \lim_{x\to 3^+} f(x) \quad \lim_{x\to 3} f(x)$$



---

4. A graph approaches the same value from both sides at x = -1, but the function value at x = -1 is 5 while the graph approaches 2. Does the limit exist? If so, what is it?

Algebraic Limits

5.

$$\lim_{x\to 5}(x^2 - 2x)$$

6.

$$\lim_{x\to -2}(3x^2 + x - 1)$$

7.

$$\lim_{x\to 1} \frac{x^2-1}{x-1}$$

8.

$$\lim_{x\to 3} \frac{x^2-9}{x-3}$$

9.

$$\lim_{x\to 2} \frac{x^2-4}{x^2-3x+2}$$

Infinite Limits

10.

$$\lim_{x\to 0^-} \frac{1}{x} \quad \lim_{x\to 0^+} \frac{1}{x}$$

11.

$$\lim_{x\to 3} \frac{1}{x-3}$$

Classification

12. Classify the discontinuity:

$$f(x) = \frac{x^2-4}{x-2}$$

13. Classify the discontinuity:

$$f(x) = \frac{1}{x-5}$$

14. Classify the discontinuity:

$$f(x) = \begin{cases} x+1 & x < 2 \\ x-3 & x \geq 2 \end{cases}$$

Justify

15. Explain why a removable discontinuity can still have a limit.

16. Explain why infinite discontinuities prevent a two-sided limit from existing.

17. Explain why jump discontinuities prevent a two-sided limit from existing.

18. Explain why factoring is necessary when evaluating limits algebraically.

Solutions

1. The graph approaches 3 from both sides.

$$\lim_{x\to 2} f(x) = 3$$

The limit exists.

2.

$$\lim_{x\to 1^-} f(x) = 2$$

$$\lim_{x\to 1^+} f(x) = 0$$

Since the one-sided limits are not equal, the two-sided limit does not exist.

3.

$$\lim_{x\to 2^-} f(x) = +\infty$$

$$\lim_{x\to 2^+} f(x) = -\infty$$

The two-sided limit does not exist.

4. The graph approaches 2 from both sides.

$$\lim_{x\to -1} f(x) = 2$$

The limit exists even though the function value is 5.

5. Direct substitution:

$$5^2 - 2(5) = 25 - 10 = 15$$

6.

$$3(-2)^2 + (-2) - 1 = 3(4) - 2 - 1 = 12 - 3 = 9$$

7. Factor numerator:

$$\frac{(x-1)(x+1)}{x-1}$$

Cancel:

$$x+1$$

Substitute:

$$2$$

8.

$$\frac{(x-3)(x+3)}{x-3}$$

Simplifies to:

$$x+3$$

Substitute:

$$6$$

9. Factor:

$$\frac{(x-2)(x+2)}{(x-1)(x-2)}$$

Cancel:

$$\frac{x+2}{x-1}$$

Substitute:

$$\frac{4}{1} = 4$$

10.

$$\lim_{x\to 0^-} \frac{1}{x} = -\infty$$

$$\lim_{x\to 0^+} \frac{1}{x} = +\infty$$

The two-sided limit does not exist.

11. As

$$x \to 3^-$$

, the function decreases without bound.

$$-\infty$$

As

$$x \to 3^+$$

, the function increases without bound.

$$+\infty$$

The limit does not exist.

12.

$$\frac{x^2-4}{x-2}$$

Factor:

$$\frac{(x-2)(x+2)}{x-2}$$

Cancel → removable discontinuity (hole at x = 2).

13.

$$\frac{1}{x-5}$$

Denominator zero at x = 5 and does not cancel. Infinite discontinuity (vertical asymptote).

14. The function changes formulas at x = 2. Left and right limits differ. Jump discontinuity.

15. A removable discontinuity occurs when a factor cancels. The surrounding values still approach a single number, so the limit exists.

16. Infinite discontinuities cause the function to increase or decrease without bound. Since the value does not approach a finite number, the two-sided limit does not exist.

17. Jump discontinuities produce unequal left and right limits. A two-sided limit requires equality, so it does not exist.

18. Factoring removes removable discontinuities by canceling common factors, allowing evaluation of the simplified expression.